

MILESTONES

A voice-based assistive communication system using SiWG917 (Edge ML + WiFi) to detect patient commands (Water, Food, Help, Emergency) and notify caregivers in real time with audio feedback and dashboard updates.

1. TIMELINE (PLANNED VS ACHIEVED)

Phase	Timeline	Status
Foundation (Setup, Dataset)	Month 1	Completed
ML Development (Training & Deployment)	Month 2	Completed
Integration (Hardware + WiFi + Dashboard)	Month 3	In Progress

2. CURRENT STATUS

- On-device ML inference working (<200 ms latency)
- Voice & button inputs functional
- Direct WiFi communication to server
- Audio feedback and dashboard updates operational

Status: Functional prototype under optimization

3. TOP RISKS

1. Wi-Fi Reliability (Primary Risk)

Possible delays or data loss in communication

Mitigation: Retry mechanisms, data buffering

2. Model Accuracy in Noisy Environments

Reduced performance due to background noise

Mitigation: Data augmentation, model tuning

3. Hardware Variability

Microphone/speaker differences affecting performance

Mitigation: Calibration and signal filtering